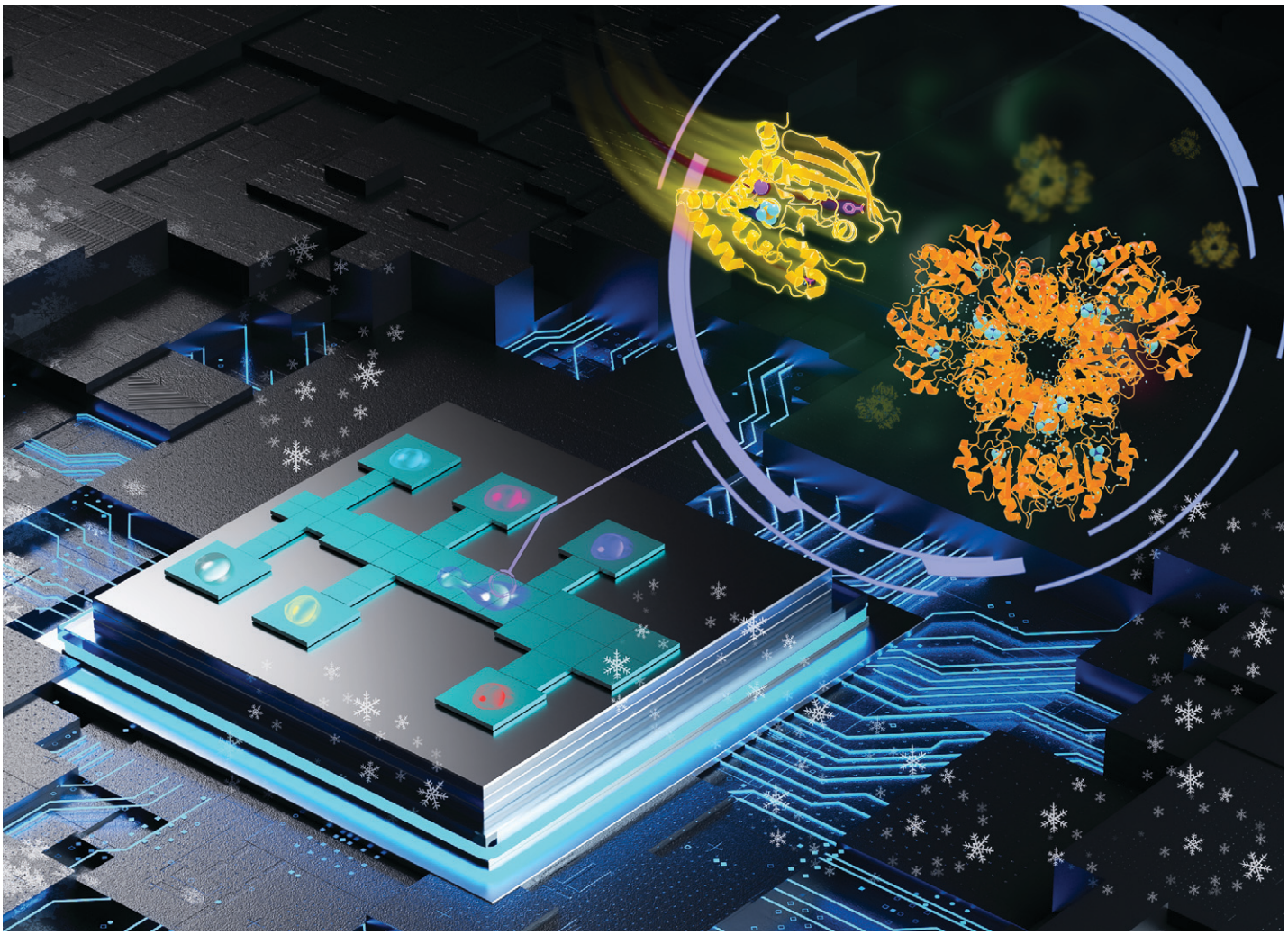

Featuring work from Professor Hui Yang's laboratory, Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, Guangdong, P. R. China and Professor Hong Chen's laboratory, Pen-Tung Sah Institute of Micro-Nano Science and Technology, Xiamen University, Fujian, P. R. China.

A low-temperature digital microfluidic system used for protein–protein interaction detection

A low-temperature digital microfluidic system (LTDMF) was developed here, which provided the essential temperature environment for protein-protein interaction (PPI) detection. The LTDMF chip was used to achieve the automated, rapid, non-invasive, and efficient PPI detection. With this merit, the LTDMF chip has the potential to facilitate PPI studies in the future.

As featured in:

See Hui Yang, Hong Chen *et al.*, *Lab Chip*, 2023, **23**, 4390.

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A low-temperature digital microfluidic system used for protein–protein interaction detection†

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The occurrence, development and prediction of various biological processes and diseases are inseparable from the protein–protein interaction (PPI), so it is extremely meaningful to perfect PPI networks. However, shortcomings of traditional detection methods, such as protein degradation, long detection time, complex operation, poor automation and high cost, restrict the rapid development of PPI networks. Here, a low-temperature digital microfluidic (LTDMF) system-based PPI detection box (LTDMF-PPI-Box) was developed to achieve rapid, lossless and efficient PPI detection. It consists of a PMMA shell, LTDMF-PPI and an integrated temperature control system. LTDMF reduces the PPI detection time from tens of hours to 1.5 hours by programmatically controlling the movement of droplets. Moreover, an integrated thermoelectric cooler (TEC) ensures an operating temperature of 4 °C, resulting in a protein protection up to 90%. The interaction between RILP protein and Rab26 protein which has a close connection to insulin secretion was demonstrated as a prototype to illustrate the feasibility of the LTDMF-PPI-Box. LTDMF with automation characteristics is capable of meeting the requirement of high-throughput screening of interacting proteins; therefore, the LTDMF-PPI-Box is expected to accelerate the establishment of the PPI network in the future.

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Introduction

With the completion of the Human Genome Project, life science has entered the post-genomic era, that is, the research of proteomics. PPI is an important approach and tool for proteomics research.[1] The occurrence, development and prediction of various biological processes and diseases are inseparable from PPI. Current methods of PPI include tandem affinity purification (TAP),[2] fluorescence colocalization,[3] fluorescence bimolecular complementation (BiFC),[4] fluorescence energy resonance transfer (FRET),[5] co-immunoprecipitation (Co-IP),[6] *etc.* However, the above methods are all labor-intensive with limited automation, and are also difficult to execute in parallel. Excitingly, protein chips[7,8] have been developed to solve some of these difficulties; they are high-throughput micro-analysis systems that usually consist of a high-density spot matrix of hundreds to thousands of different proteins on a glass slide. Upon immune reaction, the matrix of protein spot arrays can measure multiple analytes in a small number of samples at once. In addition, the ability of protein microarray technology to screen proteins in a high-throughput manner is also one of its major advantages. However, the manufacture of protein chips requires complex expertise, and their disposability after single use and high production costs hinder their wide-spread application in most laboratories.

DMF is a powerful microfluidic technology, which not only retains the advantages of small sample volume, good airtightness and fast analysis speed required by traditional microfluidic chips, but also does not rely on complex structures such as pumps and valves.[9] It has the characteristics of a high level of automation, simple internal structure and strong scalability, which has very high potential in the fields of medical diagnosis, chemical analysis and life science. In the past decade, many DMF-based platforms have been developed for automated detection, such as magnetic bead-based immunoassays,[10–12] cell culture,[13–16] protein analysis[17,18] and nucleic acid amplification.[19–22]

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Compared with the shortcomings of multi-step sample processing and cumbersome pretreatment processes in proteomics experiments, the ability of DMF to process multiple reagents at the same time makes it very suitable for this type of research. However, only few have been published since most traditional DMF platforms have obstacles of relatively large sample volumes and unadjustable temperature control for various biological samples, limiting their applications on PPI detection. Seale *et al.*[23] reported a method for immunoprecipitation by digital microfluidics (DMF-IP) for sample cleanup and targeted protein preconcentration. This method allows 80% recovery of model proteins from approximately microliter volumes of serum in a sample-to-answer run time of approximately 25 min. It also reduces operation steps and time, but it cannot meet the requirement that the protein needs to be operated at low temperature.

Here, a rapid, low-cost, automated and low-temperature PPI detection box is developed, which is based on a digital microfluidic system. It consists of a PMMA shell, LTDMF-PPI (low-temperature digital microfluidic system-based protein–protein interaction) and a temperature control system. In particular, TEC was used to decrease the working temperature from room temperature to 4 °C in 60 s and then kept stable, protecting the bioactivity of proteins. Instead of protein A/G agarose, tosyl-functionalized magnetic beads were conjugated with an antibody, and then a protein complex with a fluorescent group in the sample was captured by the magnetic beads through immune response. The interaction between proteins was then detected through a conventional fluorescence microscope. The interaction between RILP protein and Rab26 protein, which is closely associated with insulin secretion, is employed to assess the efficacy of the LTDMF-PPI-Box. Compared with conventional techniques, the LTDMF-PPI-Box uses antibody-conjugated magnetic beads to detect the interactions between different proteins, making this platform reusable and cost-effective. The automation and standardized operation of DMF can also provide a solid foundation for realizing accurate quantification in the future.

Experiment

Reagents and materials

Toluenesulfonyl-modified magnetic beads were purchased from Biomag Biotechnology Co., Ltd. (Wuxi, China). Mouse anti-Myc monoclonal antibody was purchased from Abgreen Co., Ltd. (Guangzhou, China). RIPA lysis buffer was purchased from Xiamen Wanhexin Biological Technology Co., Ltd. (Xiamen, China). A protease inhibitor was purchased from Sangon Biotech Co., Ltd. (Shanghai, China). Microchem SU8-2015 was purchased from Nippon Kayaku Co., Ltd. (USA). Quartz glass was purchased from Donghai County Weida Quartz Products Co., Ltd. (Lianyungang, China). ITO glass was purchased from Luoyang Guluo Glass Co., Ltd. (Luoyang, China). Teflon AF 1600 was purchased from Shanghai Puchun Industrial Co., Ltd. (Shanghai, China). Pluronic F127 was purchased from Sigma (USA). Silicone oil (viscosity 0–10 MPa s) was purchased from Aladdin (Shanghai, China).

The buffer solutions were prepared as follows: PBS (0.1 M, pH = 7.4); washing buffer (0.5 wt% Tween-20 in PBS, pH = 7.4); sample dilution buffer (0.1 wt% BSA in PBS); coupling buffer (0.1 M PBS, pH = 8.5); SDS loading buffer (0.1 M Tris-HCl, 4% (w/v) SDS, 0.2% (w/v) bromophenol blue, 20% glycerol, 2% (v/v) β-mercaptoethanol, pH = 6.8); lysis buffer for cells (1% (v/v) Triton X-100, 1 mM PMSF, EDTA-free proteinase inhibitor cocktail in GST binding buffer (1 : 1000)); Co-IP buffer (20 mM Hepes, 150 mM NaCl, 1 mM EDTA, 0.5% (v/v) NP-40, pH = 7.4). Blocking solution and storage buffer were purchased from Biomag Biotechnology Co., Ltd. (Wuxi, China).

Design of the LTDMF-PPI-Box

As shown in Fig. 1, the LTDMF-PPI-Box is composed of two parts: a DMF system and a temperature control system. The DMF system includes an LTDMF-PPI chip, DMF driving system, magnet and chip holder. The temperature control system includes: a TEC and control system, thermal silica gel, lithium battery and water cooling radiator (see Fig. S1A and B†). The LTDMF-PPI chip is mounted on the holder, and then controlled with a computer through the DMF driving system (see Fig. S1C†). The LTDMF-PPI chip is closely attached on the cold side of TEC (TEC1-12706, 40 mm × 40 mm) by using thermal silica gel; the hot side of TEC is attached to a water cooling radiator (see Fig. S1D and Table S1†). A cooling fan (with the red light in Fig. S1A†) is fixed at the exhaust port of the shell, which is connected with the water cooling radiator by a water pipe. The lithium battery provides a stable power supply for the DMF and temperature control systems. The shell of the LTDMF-PPI-Box is made of polymethyl methacrylate (see Fig. S2†), and more details of the system are described in the ESI.†

Design of the LTDMF-PPI chip

The LTDMF-PPI chip consists of two parts: the top plate and the bottom plate. A piece of indium tin oxide (ITO) glass is used as the top plate, and the driving electrodes are fabricated on the bottom plate by photolithography whose photomask is processed by Shenzhen Qingyi Optoelectronics Co., Ltd. (Shenzhen, China). The design of the prepared bottom plate includes: a fluid reservoir region composed of 6 reservoir electrodes (6 mm × 5 mm for each electrode), a cell lysis region composed of 12 driving electrodes (1.6 mm × 1.6 mm for each electrode), a reaction region composed of 20 driving electrodes (2.2 mm × 2.2 mm for each electrode) and a sample preparation area composed of 7 driving electrodes (3 mm × 3 mm for each electrode). The fabrication of the LTDMF-PPI chip's bottom plate is described in Fig. S3.†

Preparation of capture antibody conjugated magnetic beads

The antibody (anti-Myc) was dialyzed to remove free groups that may exist. Toluenesulfonyl magnetic beads (300 μL, 20 mg

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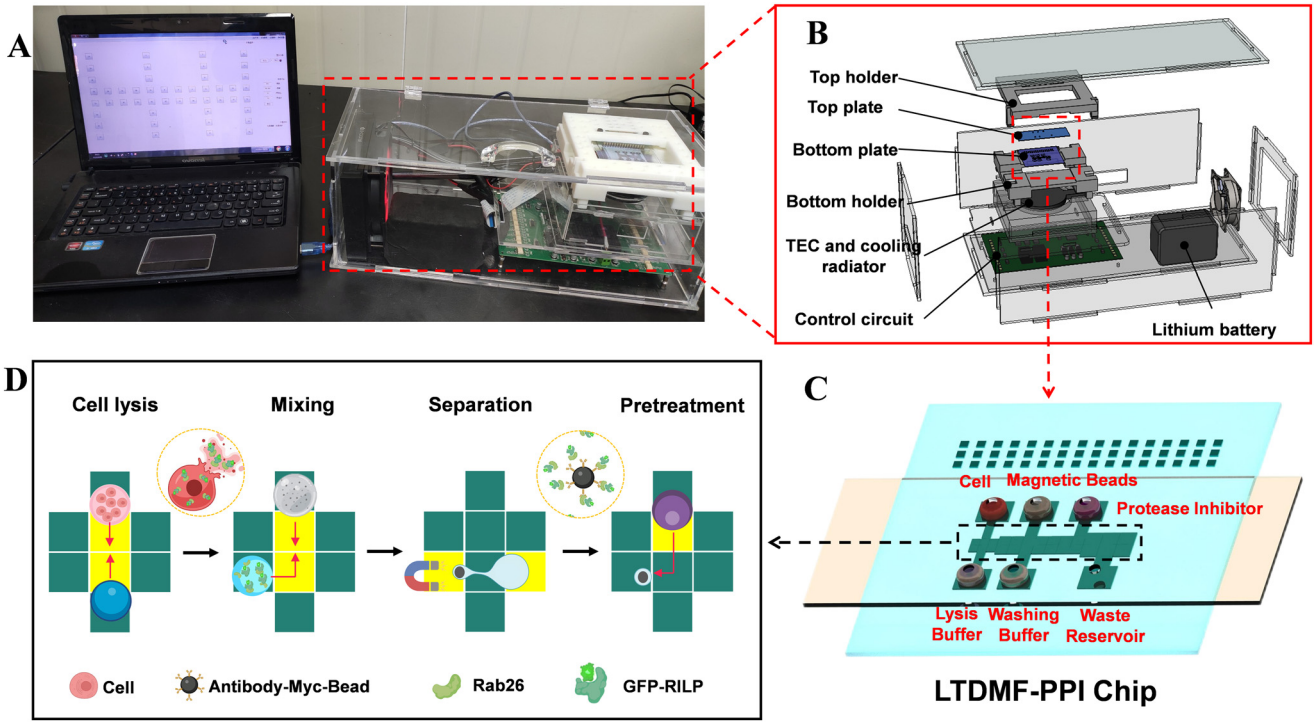

Fig. 1 LTDMF-PPI-Box used for PPI detection. (A) LTDMF-PPI-Box. (B) 3D exploded view of LTDMF-PPI-Box. (C) LTDMF-PPI chip. (D) Schematic illustration of the process of PPI on the LTDMF-PPI chip.

mL^{-1}) were washed 3 times with the coupling buffer. After removing the supernatant, 20 μL antibody (anti-Myc, 0.68 mg mL^{-1}) was mixed with the magnetic beads. The coupling buffer was then added and incubated at 4 °C for 12 h with shaking. After removing the supernatant, the blocking buffer (100 μL) was added to block the residual active sites and incubate at 4 °C for 6 h with shaking. After then, the magnetic beads were washed with 1 mL coupling buffer and washing buffer alternately. Finally, anti-Myc conjugated magnetic beads (5 mg mL^{-1}) were obtained and stored at 4 °C.

PPI detection on the LTDMF-PPI-Box

The LTDMF-PPI chip was mounted on the chip holder to ensure that the pogo pin interface was in full contact with the pad on the chip. Samples were introduced into the reservoirs of the LTDMF-PPI chip. The PPI detection process is as follows: (1) droplets with antibody-conjugated magnetic beads were generated in the LTDMF-PPI chip, and then moved to a fixed position; (2) magnetic beads in the droplet were restrained by an external magnet, and the droplet was driven away from the magnetic beads to the waste reservoir; (3) droplets containing cells and lysis buffer were generated from individual reservoirs and mixed by driving electrodes; (4) the above droplets were mixed with the magnetic beads; (5) after incubation, the magnetic beads were restrained by the external magnet and the droplet was driven to the waste reservoir; (6) droplets were generated from the washing buffer reservoir and then the magnetic beads were washed 6 times; (7) a droplet containing a protease inhibitor and PMSF was generated and mixed with the magnetic beads; (8) the droplet-containing magnetic beads were collected, sealed with silicone oil and prepared for fluorescence detection.

Results and discussion

Working principle of LTDMF-PPI

A stable temperature control system, convenient non-contact transportation, and accurate and automated operation are essential for precise PPI detection. The LTDMF-PPI platform with a temperature control system was developed here, where TEC was used to provide the required low-temperature environment. TEC has two semiconductors (N-type and P-type) connected in series, generating an endothermic process that results in a cold side when the direct current flows from the P-type material to the N-type material, and *vice versa* (see Fig. S1D†). The heat flux is proportional to the magnitude of the applied direct current, so it is possible to control the heat flux and then the temperature of the cold side by varying the input current. As shown in Fig. 1B, the LTDMF-PPI chip is mounted above and clung to the cold side of TEC directly. LTDMF-PPI chip's temperature decreases to 4 °C which meets the requirement of PPI detection. The hot side of TEC is clung to a water cooling radiator and the heat flux is transferred to water, and then the water circulating between the water cooling radiator and a cooling fan transfers the heat flux to the outside of the box.

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As shown in Fig. 1C, the LTDMF-PPI chip contains 6 reservoirs which are used to store and introduce the cell suspension, cell lysis buffer, antibody-conjugated magnetic beads, protease inhibitor, washing buffer, and waste, respectively. Fig. 1D shows the process of PPI detection: transfected 293T cell suspension is mixed with lysis buffer to release the target protein in the cell; the lysed mixture is then mixed with anti-Myc-conjugated magnetic beads, and the target protein is captured by the magnetic beads; an external magnet is applied to perform magnetic separation to remove the supernatant, and the magnetic beads are washed with a washing buffer several times; the magnetic beads are then suspended in the buffer containing the protease inhibitor; the prepared sample droplet is collected and then detected using a microscope or western blot. The working temperature of the above steps should be kept at 4 °C. For traditional methods, it is very cumbersome to meet such a requirement and the total time is from several hours to days. With the LTDMF-PPI-Box, accurate temperature control and automatic operation are achieved, and the time is decreased to 1.5 hours. It is worth noting that the above time does not include the confocal microscopy imaging. Because the confocal imaging time is closely related to the operator's operating experience, it is impossible to determine the exact detection time. Silicone oil used in the DMF chip also helps to decrease external contamination and liquid volatilization, and therefore ensures accurate results. Accordingly, with the advantages of LTDMF-PPI, the interaction among proteins can be conducted in a highly simple, quick and accurate manner.

Feasibility of temperature control in LTDMF-PPI

Using a certain cell type as a model to express the corresponding proteins for PPI detection is one of the most commonly methods. However, these proteins are extremely easy to become inactivated, so the temperature must be kept below 10 °C during the PPI detection. The most commonly used method is to place the centrifuge tube containing the proteins in ice, which is very cumbersome and time consuming. Moreover, the ice needs to be replaced in time. When the centrifuge tube is picked up to perform various operations such as adding solution, shaking and separating, the proteins will be warmed up and result in denaturation. In this work, the above problems were solved by combining the LTDMF-PPI chip with the temperature control system in the LTDMF-PPI-Box. The temperature control system is used to keep the working temperature at about 4 °C, and the LTDMF-PPI-Box performs all operations quickly and automatically, therefore facilitating PPI detection.

As shown in Fig. 2A, a thermal imaging system and finite element analysis simulation were used to characterize the performance of the temperature control system. The top row in Fig. 2A shows the simulation results of temperature distribution on the TEC's surface. At the beginning, the temperature at the TEC's surface was room temperature (20 °C). After cooling for 30 s and 60 s, the temperature at the working area decreased to 12 °C and 4 °C, respectively. The bottom row in Fig. 2A shows the results from the thermal imaging system. At the beginning, the temperature at the TEC's surface was 20.0 °C. After cooling for 30 s, it decreased to 12.6 °C, and then achieved 3.3 °C after another 30 s. Under three kinds of room temperature conditions, we tested the transient response, dynamic response and stability of the low temperature control system, respectively. Fig. S4A† shows that the system can drop and stabilize the temperature of the working area to 4 °C within 60 s. Fig. S4B† demonstrates the system's excellent adaptability and stability when subjected to 10 cycles of varying currents. The results of simulation and thermal imaging both show that the temperature of the working area can achieve 4 °C after cooling for 60 s, and the temperature distribution shows good uniformity. As shown in Fig. 2B, by comparing the uniformity of the simulated temperature and real temperature of TEC, we found that although the edge temperature of TEC was relatively high, the temperature in the core area was around 4 °C, which was enough to cover the droplet movement area.

The working principle of the DMF chip is predicated on the alteration of the droplet contact angle. When the initial contact angle of the droplet is larger, the motion of the droplet will be more fluid, otherwise it will be more difficult for the droplet to move. It has been reported that temperature can affect the hydrophobicity of the material surface. Hence, we tested the contact angle variation of the droplet on a Teflon hydrophobic layer at diverse temperatures. As demonstrated in Fig. 2C, the initial contact angle of the droplet increased steadily with the reduction of temperature, which was in agreement with the results in the literature.[24] This transformation is advantageous for the movement of droplets at low temperature. Conversely, when scientists develop some techniques that necessitate high temperature, such as PCR, *etc.*, based on DMF, it is essential to suitably increase the control voltage to maintain the controllability of droplets.

The function area of the LTDMF-PPI chip contains electrodes with three different sizes (cell lysis region, reaction region and sample preparation region), and the droplets' volume has significant effects on the reaction and results. In Fig. 2D, reproducibility on the volume of droplets generated from such three regions was characterized respectively. 20 droplets were subsequently generated from the reservoirs in the three regions, and then the volume of the generated droplets was calculated by their areas and the heights between the two plates (400 μm). As shown in Fig. 2D, CV (%) on the volume of the droplets is 2.19% for the cell lysis region, 2.67% for the reaction region and 2.56% for the sample preparation region. These CV values are comparable with those generated by manual pipetting, and enough for conventional use.

Optimization of LTDMF-PPI

Cell lysis is an essential step for PPI detection on the LTDMF-PPI chip. The reagent's consumption on the LTDMF-

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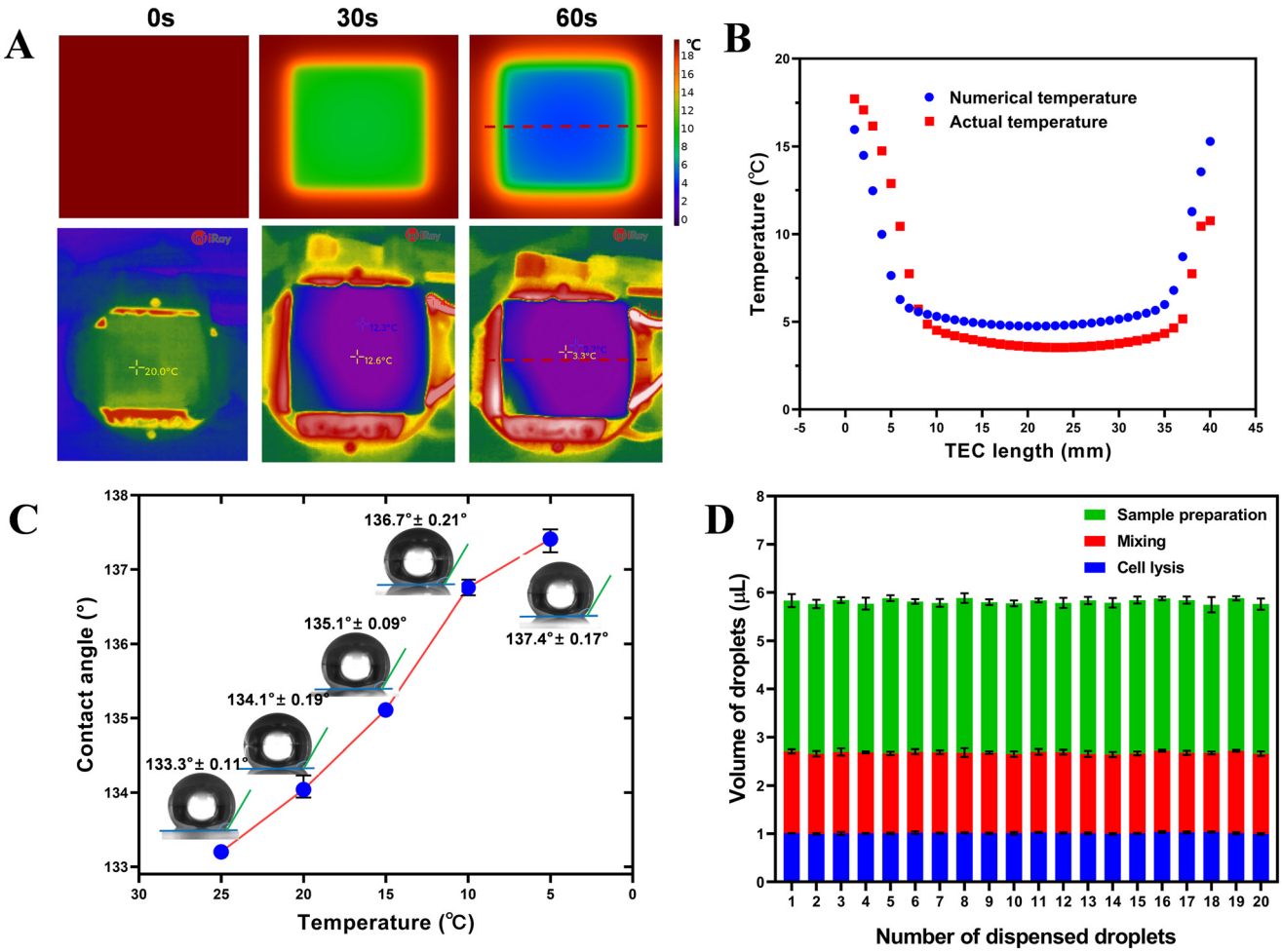

Fig. 2 (A) Simulation results (top row) and thermal images (bottom row) of the temperature control system (temperature distribution on the TEC's surface after being cooled for different time durations). (B) The comparison of the uniformity of the simulated temperature and real temperature. (C) The curve of the droplet contact angle changing with temperature. (D) Volume uniformity of the droplets generated from the reservoirs in three regions. 20 droplets are generated from each reservoir; CVs are 2.19% (cell lysis region), 2.67% (reaction region) and 2.56% (sample preparation region), respectively.

PPI chip is very small (only 1–3 μL), and parameters optimized in conventional methods cannot be fully applied for the chip, so it is necessary to validate and optimize the parameters which are suitable for the LTDMF-PPI chip. The cells (293T, about 1×10^6 cells per mL) were lysed by a conventional method and the LTDMF-PPI chip (tube and DMF) respectively, and then the protein's concentration was measured using a BCA kit to determine the best lysis time (see Fig. 3A). For the conventional method, lysing efficiency increases quickly and then becomes stable in 10 min. For the LTDMF-PPI chip, the efficiency's increase is slower than that of the conventional method, and it also becomes stable after 15 min. The lysate's volume ratio used in the conventional method (10 : 1) is much larger than that used in the LTDMF-PPI chip (1 : 1), so it exhibits stronger lysis ability in the early stage of lysis (<10 min). But after 15 min, the efficiencies for the conventional method and LTDMF-PPI chip are almost the same. In addition, the lysis efficiencies for various cells were also compared (see Fig. S5†). These cells (U89, K562 and 293T) are chosen because they are often used in cell transfection. The results show that when the suspension is mixed with a lysis buffer for 30 minutes, the cells are completely lysed by either the conventional method or the LTDMF-PPI chip.

In the Co-IP experiment, the detection time is generally more than 8 h because of a large volume of solution. When the Co-IP experiment is performed on the LTDMF-PPI chip, the reaction happens inside the droplets of a very small volume (~ several microliters). The incubation time dramatically decreases due to fast diffusion and reaction inside the droplet. The required incubation time for the LTDMF-PPI chip is investigated here. As shown in Fig. 3B, incubation efficiency increases with increasing incubating time, and then becomes stable after 30 min. This means that almost all protein complexes in the droplet have been captured by magnetic beads. In the following studies, an incubation time of 30 min was adopted.

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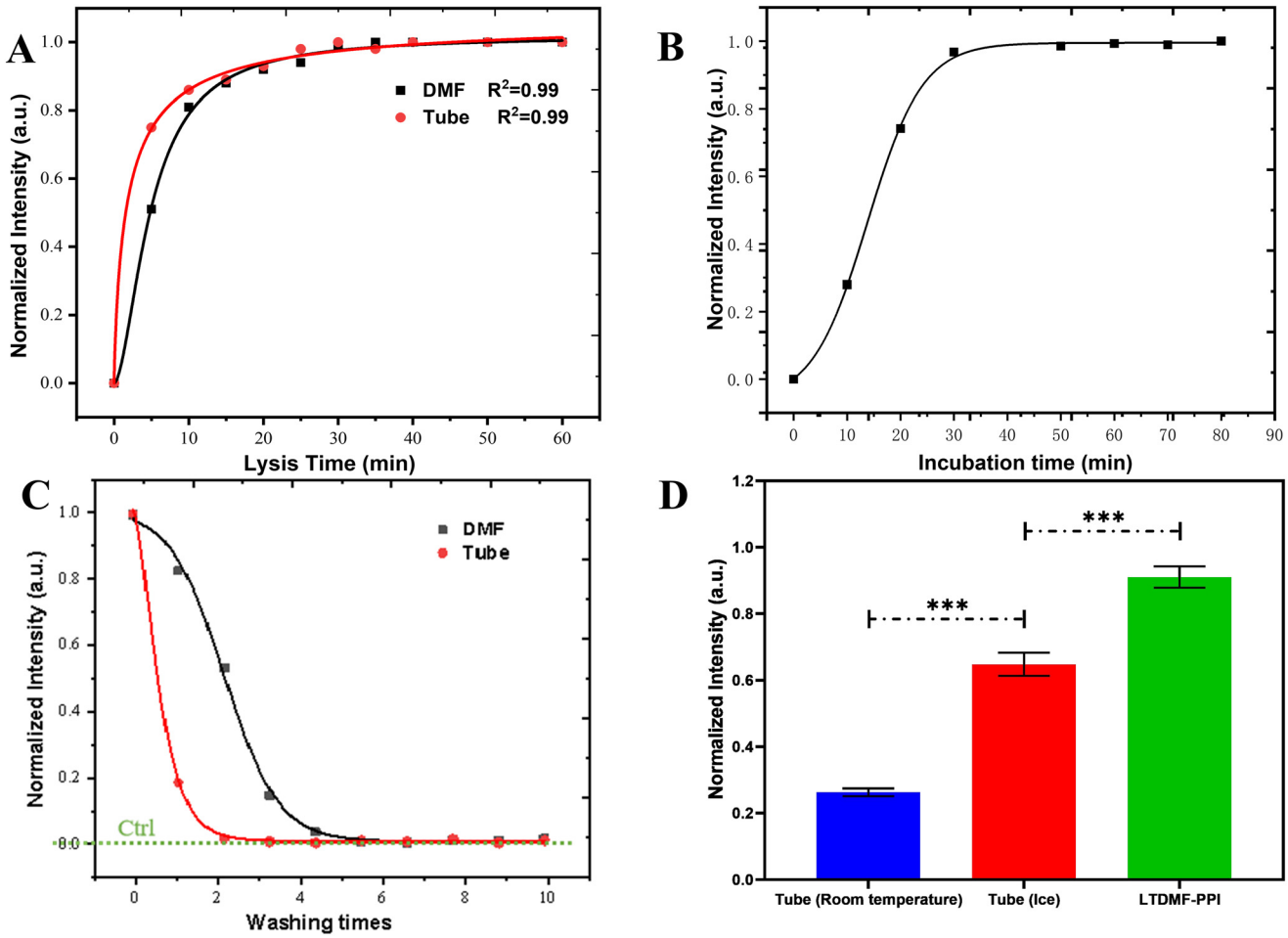

Fig. 3 Optimization of PPI detection on the LTDMF-PPI chip. (A) Efficiency of cell lysis. (B) Incubation time. (C) Washing efficiency. (D) Protein survival rates under different temperature conditions (NS: not significant, $p > 0.1$; $***p < 0.001$).

The magnetic beads are subjected to multiple wash cycles to eliminate unbound complexes after incubation and the washing solutions after each washing step are carefully collected in a 96-well plate. A BCA kit was used to measure the protein content in the collected solutions, and the results are shown in Fig. 3C. For the conventional method, the signal decreased dramatically and reached the background after the second washing. For the LTDMF-PPI chip, the signal decreased slower than that of the conventional method, and reached the background after the fifth washing. The results showed that washing 4 or 5 times is enough to remove almost all the unbound protein complexes from the LTDMF-PPI chip.

In order to investigate the influence of the LTDMF-PPI-Box on the protein retention rate, we performed conventional methods at room temperature and in ice, and quantified the proteins obtained using the BCA kit. The results shown in Fig. 3D indicate that at room temperature, the protein survival rate was only 25%, while that in the ice block was approximately 60%, both of which were far lower than the 90% retention rate, demonstrating the excellent performance of LTDMF-PPI.

LTDMF-PPI used for PPI detection

Disturbance in insulin secretion is closely related to occurrence and development of diabetes. The secretion process of insulin is strictly controlled by vesicle transport. RILP protein has become the main integrator of the transport in the body through the interaction with a variety of Rab proteins (such as Rab7, Rab34, Rab36 and Rab24).[25–28] The interaction between RILP protein and Rab7 protein is defined as an indispensable tool to promote lysosomal degradation of proinsulin, but the mechanism of how RILP guides insulin particles to the lysosomal degradation pathway is not fully understood.[29] Current studies have shown that Rab26 protein is a secretory vesicle protein and participates in the secretion of starch granules and neurosynaptic vesicles.[30,31] So is there an interaction between Rab26 and RILP, which in turn is related to insulin particles?

Co-IP is a classical method to study protein interactions, and is based on specific binding between antibodies and antigens, and it is also the most convincing method in routine experiments. The beads used in the traditional Co-IP are protein

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A/G agarose (Thermo Fisher Scientific), which are not magnetic and cannot be used in LTDMF-PPI chips. Therefore, tosyl magnetic beads were used for immunoprecipitation experiments in this report. Furthermore, we evaluated the feature change of the magnetic beads during the PPI detection. As shown in Fig. 4A, the zeta potential of the magnetic beads increases from negative to 0 after conjugation of capture antibody (anti-Myc), which should be attributed to neutralization of antibody to beads. The zeta potential of the beads decreases from 0 to negative after the protein complex's capture (GFP-RILP–Myc-Rab26), which should be attributed to the large number of protein complexes with negative charges. Meanwhile, the magnetic beads' size increases after the conjugation with the capture antibody, and then continues increasing after capturing the protein complex (see Fig. 4B). The magnetic beads' composition was analyzed by energy dispersive spectrometry (EDS) mapping and an overlay of elemental mapping images by scanning electron microscopy (SEM). The elemental mapping images before (top row in Fig. 4C) and after (middle and bottom rows in Fig. 4C) the capture further confirm that the protein complex has been successfully captured by the magnetic beads. In addition, western blot was also used to determine whether the protein complex was captured on the magnetic beads.

In order to verify the interaction between RILP and Rab26, 293T cells transfected with Myc-Rab26 and GFP-RILP plasmid vectors were used as the experimental group, and the 293T cells transfected with Myc-Vector and GFP-Vector were used as the control group. Fig. 5B shows the cells transfected with GFP-PILP or GFP-Vector. The fluorescence image of the control group (top row in Fig. 5B) shows that GFP is randomly distributed in the cell's cytoplasm and nucleus. While in the fluorescence image of the experimental group (bottom row in Fig. 5B), there is a black core inside the cell and GFP only distributed in the cytoplasm. Comparing with the reported RILP morphology,[32] it showed that GFP-RILP had successfully been expressed in the cells. Then, the PPI was performed on the LTDMF-PPI-Box, whose process is shown in Fig. 5A. The droplets containing 293T and lysis buffer were generated from the corresponding reservoirs, and then mixed to release the protein complex from the cell. The mixture was then moved to the reaction area and mixed with

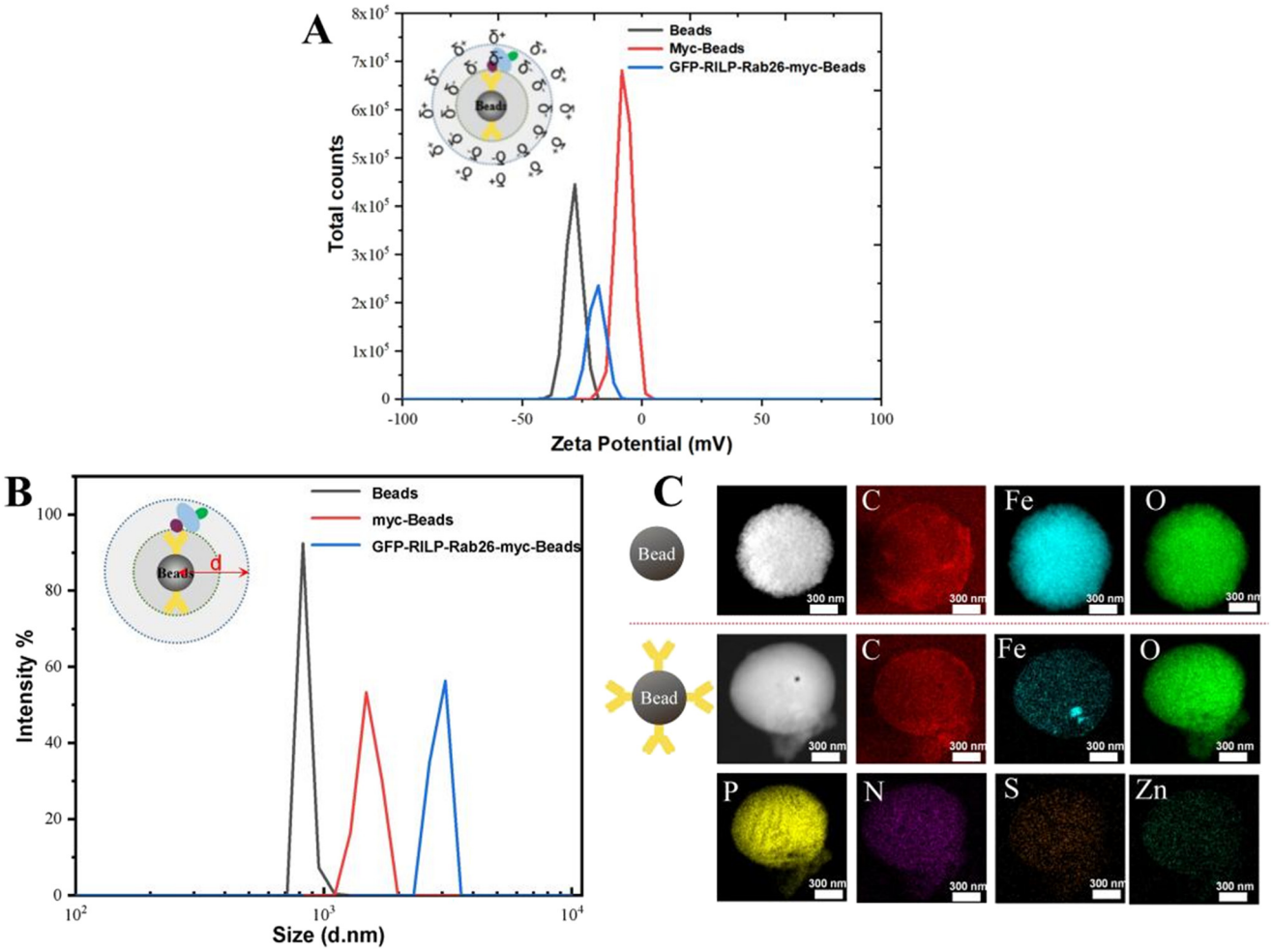

Fig. 4 Zeta potentials (A) and size distributions (B) of the magnetic beads during PPI detection. (C) SEM images and EDS mapping images of magnetic beads before (top row) and after (middle and bottom rows) the capture.

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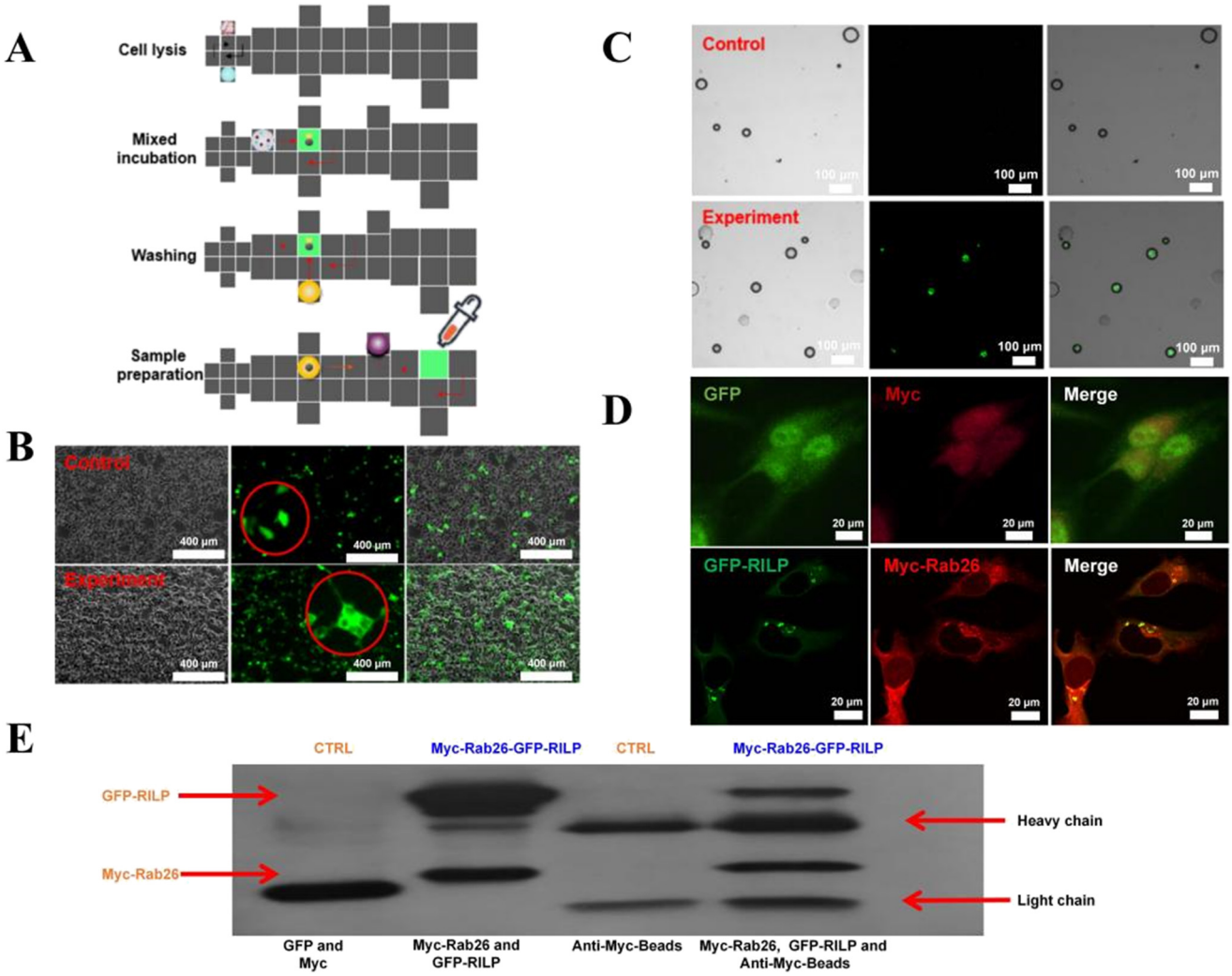

Fig. 5 PPI detection on the LTDMF-PPI chip. (A) Operation flow of the LTDMF-PPI chip for PPI detection. Droplets generated from the cell suspension and lysis buffer are mixed together to release a protein complex, then incubated with magnetic beads to capture the protein complex; after washing several times, the droplet is then mixed with a protein inhibitor, and collected for the following imaging step. (B) Images showing the interaction between Myc-Rab26 and GFP-RILP in 293T cells by a traditional method. Left column: Images from bright field; middle column: fluorescence images; right column: overlapped images. (C) Images of the interaction between Myc-Rab26 and GFP-RILP protein on the LTDMF-PPI chip. Left column: Images from bright field; middle column: fluorescence images; right column: overlapped images. (D) Co-localization of intracellular interacting proteins observed under confocal fluorescence microscopy. (E) Co-immunoprecipitation on the interaction between Rab26 and RILP.

the capture antibody-conjugated magnetic beads. During incubation, the protein complex was captured by the capture antibody on the magnetic beads. After being washed 6 times, the magnetic beads were moved to the sample preparation

Table 1 Comparison between traditional detection methods and LTDMF-PPI-Box (※ represents the advantage)

	Traditional detection methods	LTDMF-PPI-Box
Solid carrier	Protein A/G (5000 rmb per mL)	Tosyl magnetic beads※ (30 rmb per mL)
Cooling source	Ice	Thermoelectric cooler (TEC)
Cell lysis time	30 min	30 min
Operation method	Manual (≈ 2 h)	Programmed (30 min)※
Sample consumption	>100 μL	1–5 μL※
Incubation time	6–8 h	30 min※
Integration	No	LTDMF-PPI-Box※ (450 cm × 300 cm × 200 cm)
Full detection time	≥8 h	1.5 h※

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area to mix with the droplet containing protein inhibitors. After that, the mixture was collected for the following detection. Finally, a confocal microscope was used to analyze the results. The images of the control group (top row in Fig. 5C) shows that the GFP-RILP protein does not directly bind to the magnetic beads. While the images of the experimental group (bottom row in Fig. 5C) shows green fluorescence, indicating that the anti-Myc conjugated magnetic beads captured the protein complex (Myc-Rab26–GFP-RILP). Images of the interaction between Myc-Rab26 and GFP-RILP protein from multiple replicate experiments are shown in Fig. S6.† The result proves that there is an interaction between Myc-Rab26 and GFP-RILP. The co-localization of intracellular interacting proteins was observed under confocal fluorescence microscopy (see Fig. 5D). And co-immunoprecipitation also confirms the interaction between Rab26 and RILP (see Fig. 5E). Compared with conventional methods, the LTDMF-PPI-Box with temperature control can reduce interference of personnel to the greatest extent, and greatly shorten the time of the entire inspection process (see Table 1).

Conclusion

PPI is an important approach and tool for proteomics research, and conventional methods have disadvantages of long detection processes, the need for professionals, low automation, large reagent consumption, *etc.* In this paper, a portable, low-cost, automated and temperature adjustable LTDMF-PPI-Box was developed. TEC is mounted under the LTDMF-PPI chip to provide the temperature environment which is required for PPI detection. The LTDMF-PPI chip is used to achieve automation and high-efficiency operation of PPI detection, where the time is decreased to 1.5 h (excluding confocal imaging steps). The interaction between Rab26 protein and RILP protein in 293T cells, which is related to insulin secretion, is chosen as a model to show the feasibility of the LTDMF-PPI-Box. With the above merits, the LTDMF-PPI-Box has potential to facilitate PPI studies in the future.

Conflicts of interest

There are no conflicts to declare.

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